Clearance & Elimination Practice 3

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

Cefprozil, 500 mg oral every 12 hours for 10 days, has been prescribed for patient, KL. KL is a 42 year old male, who is 178 cm tall and weighs 73 kg and has a serum creatinine of 0.82 mg/dL. Cefprozil is well absorbed with a bioavailability of about 95%. The half-life of elimination in adults with normal hepatic and renal function is about 1.3 hours, 5.2 hours in adults with renal impairment (120 > CrCl > 30mL/min), and 5.9 hours in patients with renal failure (CrCl< 30mL/min),. The apparent volume of distribution is 0.23 L/kg. About 60% of the drug is excreted in the urine as unchanged drug. According to the manufacturer's labeling for adults, no dosage adjustment is necessary if the CrCl ≥30 mL/minute and the dose should be reduced by 50% if CrCl <30 mL/minute.

a Estimate KL’s creatinine clearance.

b Estimate the total body clearance (mL/min) of cefprozil in KL.

c Estimate the renal clearance (mL/min) of cefprozil in KL.

d Estimate the hepatic clearance (mL/min) of cefprozil in KL.

e If KL developed some severe complications and his serum creatinine sky-rocketed to 2.8 mg/dL, revise your estimate of KL’s creatinine clearance.

f Provide a new estimate of the total body clearance (mL/min) of cefprozil in KL assuming his volume of distribution is unchanged.

g Provide a new estimate of the renal clearance (mL/min) of cefprozil in KL.

h Provide a new estimate of the hepatic clearance (mL/min) of cefprozil in KL.